

Rigorous Step-by-Step Derivation: Orthogonalized Sieve-EL (Version 2.0 AR1) vs. Cai et al. (2026)

Baseline Framework (Cai et al., 2026)	Proposed Orthogonalized Sieve-EL (Version 2.0 AR1)
Step 1: Model Specification & DGP	
Data-Generating Process:	Data-Generating Process:
$Y_t = \alpha + \beta X_{t-1} + U_t \quad (1)$	$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t \quad (4)$
$X_t = \theta + \rho X_{t-1} + \varepsilon_t \quad (2)$	$X_t = \theta + \rho X_{t-1} + \varepsilon_t \quad (5)$
$U_t = \phi \varepsilon_t + z_t \quad (\text{Stambaugh Bias}) \quad (3)$	$U_t = \phi \varepsilon_t + z_t \quad (\text{Stambaugh Bias}) \quad (6)$
Nuisance Component: The fixed scalar intercept α . Nuisance Basis Space: $\text{span}\{1\}$.	Nuisance Component: The unknown infinite-dimensional non-linear function $m(\cdot)$. Nuisance Basis Space: $\text{span}\{P_K(W_{t-1})\}$ where $P_K \in \mathbb{R}^K$.
Step 2: Nuisance Estimation	
Because $\mathbb{E}[U_t X_{t-1}] \neq 0$ under high persistence, simple OLS is biased. They apply <i>residual addition</i> . 1. Regress X_t on X_{t-1} to get $\hat{\varepsilon}_t$:	To handle Stambaugh bias and omit variable bias simultaneously, residual addition is utilized alongside sieve approximation. 1. AR(1) regression to get $\hat{\varepsilon}_t$:
$\hat{\varepsilon}_t = X_t - \hat{\theta} - \hat{\rho} X_{t-1} \quad (7)$	$\hat{\varepsilon}_t = X_t - \hat{\theta} - \hat{\rho} X_{t-1} \quad (10)$
2. Estimate intercept via multivariate OLS:	2. Joint Sieve-Cai-Wang Estimation via OLS:
$Y_t = \tilde{\alpha} + \tilde{\beta} X_{t-1} + \tilde{\phi} \hat{\varepsilon}_t + \text{error} \quad (8)$	$Y_t = P_K(W_{t-1})' \mathbf{c} + \beta X_{t-1} + \phi \hat{\varepsilon}_t + \text{error} \quad (11)$
3. Adjust response:	This joint regression purges both omitted variable bias and endogeneity bias from $\hat{\mathbf{c}}$. 3. Adjust response:
$\tilde{Y}_t = Y_t - \tilde{\alpha} \quad (9)$	$\tilde{Y}_t = Y_t - P_K(W_{t-1})' \hat{\mathbf{c}} \quad (12)$
Step 3: Weight Orthogonalization	

To prevent the estimation error $(\tilde{\alpha} - \alpha)$ from distorting the EL limit, project raw weight $w(X_{t-1})$ onto the nuisance basis (the constant 1) via OLS:

$$\bar{w}_T = \left(\sum_{t=1}^T 1 \cdot 1 \right)^{-1} \sum_{t=1}^T 1 \cdot w(X_{t-1}) = \frac{1}{T} \sum_{t=1}^T w(X_{t-1}) \quad (13)$$

The centered weight w_t^c is the OLS residual:

$$w_t^c = w(X_{t-1}) - \bar{w}_T \quad (14)$$

By OLS normal equations (First-Order Conditions):

$$\sum_{t=1}^T 1 \cdot w_t^c \equiv 0 \quad (\text{Exact in-sample}) \quad (15)$$

To prevent the remaining sieve estimation error from distorting the EL limit, project raw weight $w(X_{t-1})$ onto the nuisance basis $(\mathbf{P}_K(W_{t-1}))$ via OLS:

$$\hat{\gamma} = \left(\sum_{t=1}^T \mathbf{P}_K \mathbf{P}_K' \right)^{-1} \sum_{t=1}^T \mathbf{P}_K w(X_{t-1}) \quad (16)$$

The centered weight w_t^c is the OLS residual:

$$w_t^c = w(X_{t-1}) - \mathbf{P}_K(W_{t-1})' \hat{\gamma} \quad (17)$$

By OLS normal equations (First-Order Conditions):

$$\sum_{t=1}^T \mathbf{P}_K(W_{t-1}) w_t^c \equiv \mathbf{0} \quad (\text{Exact in-sample}) \quad (18)$$

Step 4: Algebraic Elimination in EL Moment Condition

Construct EL moment under $H_0 : \beta = \beta_0$:

$$Z_t(\beta_0) = \left(\tilde{Y}_t - \beta_0 X_{t-1} \right) w_t^c \quad (19)$$

Expand $\tilde{Y}_t$:

$$Z_t(\beta_0) = (Y_t - \tilde{\alpha} - \beta_0 X_{t-1}) w_t^c \quad (20)$$

Substitute true DGP under null ($Y_t - \beta_0 X_{t-1} = \alpha + U_t$):

$$Z_t(\beta_0) = (\alpha + U_t - \tilde{\alpha}) w_t^c = U_t w_t^c - (\tilde{\alpha} - \alpha) w_t^c \quad (21)$$

Sum the moment conditions:

$$\sum_{t=1}^T Z_t(\beta_0) = \sum_{t=1}^T U_t w_t^c - (\tilde{\alpha} - \alpha) \underbrace{\sum_{t=1}^T w_t^c}_{=0} \quad (22)$$

Because the orthogonalization ensures $\sum w_t^c \equiv 0$, the estimation error is perfectly multiplied by zero:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t(\beta_0) = \frac{1}{\sqrt{T}} \sum_{t=1}^T U_t w_t^c \xrightarrow{d} N(0, V) \quad (23)$$

Construct EL moment under $H_0 : \beta = \beta_0$:

$$Z_t(\beta_0) = \left(\tilde{Y}_t - \beta_0 X_{t-1} \right) w_t^c \quad (24)$$

Because $\hat{\mathbf{c}}$ was estimated using the full joint regression, we have $\tilde{Y}_t - \beta_0 X_{t-1} \approx m(W_{t-1}) + U_t - \mathbf{P}_K(W_{t-1})' \mathbf{c}$. Substituting DGP and decomposition $m(W) = \mathbf{P}_K' \mathbf{c} + r_t$:

$$Z_t(\beta_0) \approx (\mathbf{P}_K' \mathbf{c} + r_t + U_t - \mathbf{P}_K' \hat{\mathbf{c}}) w_t^c \quad (25)$$

$$= U_t w_t^c + r_t w_t^c - (\hat{\mathbf{c}} - \mathbf{c})' \mathbf{P}_K w_t^c \quad (26)$$

Sum the moment conditions:

$$\sum_{t=1}^T Z_t(\beta_0) \approx \sum_{t=1}^T (U_t + r_t) w_t^c - (\hat{\mathbf{c}} - \mathbf{c})' \underbrace{\sum_{t=1}^T \mathbf{P}_K w_t^c}_{=\mathbf{0}} \quad (27)$$

Because the sieve projection ensures $\sum \mathbf{P}_K w_t^c \equiv \mathbf{0}$, the multi-dimensional parameter estimation error is multiplied by zero:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t(\beta_0) \approx \frac{1}{\sqrt{T}} \sum_{t=1}^T U_t w_t^c + \underbrace{\frac{1}{\sqrt{T}} \sum_{t=1}^T r_t w_t^c}_{=o_p(1)} \quad (28)$$

where the sieve approximation bias r_t vanishes strictly if $T^{\frac{1}{2s}} \ll K \ll T^{\frac{1}{2}}$.